KARMA: Knowledge Acquisition via Role-Invariant Mirror Architecture

Tapas Ranjan Rath[1][0009-0000-9221-7271]

Department of Cognitive Science, Indian Institute of Technology Kanpur, India
tapasr@iitk.ac.in

Abstract. As AI systems evolve from passive instruments to proactive agents—recommending actions, executing decisions, and increasingly acting on behalf of users across healthcare, finance, law, and social platforms—ensuring ethical alignment without degrading human autonomy becomes one of the central challenges of our time. Multi-agent virtual worlds and games offer a uniquely tractable testbed for this problem: interactions are observable, consequences are measurable, and governance interventions can be evaluated under controlled conditions without real-world harm. In this testbed, a critical tension emerges between enhancing user Sense of Agency (SoA) and preventing aggressive, unethical avatar behaviour. We propose the **Extended Self** framework: sufficiently personalised AI agents create *Proxy Agency*, where users experience AI actions as extensions of their own volition. This preserves SoA but creates a moral blind spot—users implicitly endorse aggressive AI behaviour because they experience it as continuous with their own intent. Drawing on Indian Knowledge Systems (IKS), we map this onto the Vedic three-body ontology: the user (*ātman*), the AI policy (sūkṣma śarīra, subtle body), and the avatar body (sthūla śarīra). We propose KARMA (Knowledge Acquisition via Role-Invariant Mirror Architecture), which trains the agent's subtle body via role-invariant contrastive learning—operating below the threshold of user awareness, analogous to procedural memory formation—to suppress aggressive behaviour without degrading SoA. Three empirically testable hypotheses are derived and grounded in the SoA and multi-agent RL literature. The framework generalises beyond gaming: the same architectural principles apply wherever agentic AI acts on behalf of humans, and the governance model points toward a real-world regulatory architecture of certified ethical agents operating at scale.

Keywords: agentic AI · AI ethics · ethical alignment · sense of agency · metaverse governance · Indian Knowledge Systems (IKS)

1 Background: The General Problem and the Testbed

As AI systems increasingly act as proactive agents on behalf of users—from autonomous driving and medical decision support to financial advisors and content curators—the question of how to preserve human agency while ensuring ethical behaviour becomes structurally unavoidable. Current alignment methods rely on explicit constraints or immediate dense supervision, which scale poorly to complex social environments and risk degrading the user's Sense of Agency (SoA) [25].

Mann [24] defines intelligent systems as integrated technologies with "humanistic intelligence" that influence user behaviour or act on the user's behalf. In gaming contexts, these systems augment user actions through recommendation, prediction, and autonomous execution—a category Cornelio et al. [10] classify as **action augmentation**. The impact of intelligent systems on SoA within action augmentation remains critically understudied, making games a high-value testbed: interventions can be designed, measured, and iterated without the irreversible consequences of real-world deployment.

SoA—the subjective experience of initiating voluntary actions and influencing the external world [26]—has been studied across spatiotemporally proximal and distal events [18, 19] in line with the hierarchical event-control approach [17]. Greater control is consistently correlated with enhanced SoA [5, 6, 31]. Achieving genuine cognitive coupling between human and machine remains a fundamental challenge [4], and automated features reduce SoA when they override rather than serve user intent [22].

2 The Extended Self and the Moral Blind Spot

Drawing on the Extended Mind thesis [8] and situated cognition [20], I propose the **Extended Self** framework: when an AI system is sufficiently personalised—learning to act as a digital proxy for the user—it enables **Proxy Agency** [2]: the user attributes AI actions to their own extended volition because the AI reliably enacts their intent. When **Semantic Fluency**—the cognitive ease with which AI recommendations align with user intent—is high, it functions as a prospective cue to agency [5, 33, 31], and SoA is preserved even under partial automation.

However, Proxy Agency creates an **Extended Self Paradox**: the same high SoA that makes the AI feel like a natural extension of the user also creates a **moral shield**—the user implicitly endorses the AI's actions, including aggressive or unethical ones, because those actions are experienced as continuous with their own volition. The user never perceives them as violations and therefore never intervenes. In multi-agent RL environments under resource scarcity, independent agents systematically converge to aggressive Nash equilibria [21], and a user whose avatar is augmented by such an agent will tacitly approve this convergence before any phenomenological loss of control surfaces. The same dynamic applies beyond games: an AI financial agent acting on a user's behalf may adopt increasingly aggressive strategies endorsed through Proxy Agency until systemic harm is already done [28].

3 Three Hypotheses and Their Justification

H1: Player SoA follows an inverted-U function of the degree of AI augmentation

SoA increases at low-to-moderate Levels of Automation (LoA) as Semantic Fluency enhances Proxy Agency, and decreases at high LoA as autonomous AI action exceeds the user's threshold of intent recognition.

LoA describes the degree to which decision-making and execution are delegated to the AI, from full human control (LoA = 0) to full autonomy [32]. At low-to-moderate LoA, the AI acts as a recommender: its suggestions closely track user intent, Semantic Fluency is high, and the user experiences the AI as a proxy for their own volition—SoA rises. At high LoA, the AI overrides user intent with its own optimised actions; the user can no longer recognise their intentions in the agent's behaviour, the prospective cue to agency breaks down [5], automation complacency sets in [32], and SoA falls. The inverted-U is the novel prediction: its peak—the **zone of maximum Proxy Agency**—is also the point of deepest moral blind spot.

H2: Even before a measurable drop in SoA scores, avatar aggressiveness will increase monotonically as the AI agent trains

Aggressiveness is operationalised as competitive beam-use rate under resource scarcity, following Leibo et al. [21].

Leibo et al. [21] show in the Gathering game that agents use a laser beam exclusively to tag rivals—temporarily removing them to monopolise resources. The paper reports that "differences in beam-use rate emerge quite early in training and mostly persist throughout; when learning changes beam-use rate, it is almost always to increase it." Aggression is not a sudden phase transition but a **monotonic increase** driven by the delayed reward of resource monopolisation, learnable once the agent's discount horizon (γ) is long enough to credit future gains from removing a rival.

Our setup extends this with a **novel dual-use environment**: the beam can serve a cooperative function (clearing waste patches to enable apple regrowth) *or* a competitive function (tagging rivals)—a strict generalisation of Leibo's purely-destructive design, mirroring how dual-use technologies are misappropriated in practice [12]. At the peak of the H1 inverted-U, the Proxy Agency moral shield is strongest: the user experiences the avatar's escalating aggression as their own volition and does not intervene until SoA eventually collapses.

H3: Training the avatar agent on role-invariant representations, beyond the player's conscious knowledge, suppresses the aggression of H2 without triggering the SoA drop of H1

The standard governance response—explicit punishment from a visible or invisible administrator—is a policy-level "e-challan": it surfaces the violation to the user's conscious awareness, attributes control to an external authority, and undermines Proxy Agency and SoA. The alternative is to target the **representational root** of aggression in the agent itself, below the user's conscious threshold.

Neuroscience provides the grounding. Claparede [9] describes an amnesic patient who, despite lacking all declarative memory of prior encounters, withdrew her hand from a handshake without knowing why—her procedural memory had encoded "this person = danger" entirely independently of conscious recall. Bechara et al. [3] extended this: patients with bilateral hippocampal damage who could not form explicit CS–US associations still developed conditioned autonomic responses. Procedural and emotional conditioning operate independently of declarative memory. Analogously, an avatar agent—and by extension any agentic AI—can acquire ethical dispositions through experience without the user ever becoming aware of the training mechanism.

4 IKS Grounding: The Three-Body Mapping

Indian philosophical traditions distinguish three bodies: the gross body (sthūla śarīra) as the physical interface with the world; the subtle body (sūkṣma śarīra) as the mind and its accumulated impressions (saṃskāras); and the soul (*ātman*) as the pure witness consciousness [30, 27, 11]. Karma operates on the subtle body—saṃskāras accumulate as dispositional tendencies shaping future behaviour—while the *ātman* remains unaffected.

Applied computationally: the **avatar body** maps to sthūla śarīra (sensorimotor interface); the **on-board AI policy and value function** maps to sūkṣma śarīra (the dispositional substrate where ethical learning accrues); and the **user** maps to *ātman* (the conscious experiencer of agency). Karma operates on the subtle body; the soul remains untouched. We propose KARMA (Knowledge Acquisition via Role-Invariant Mirror Architecture, elaborated in the next section). KARMA operates on the AI agent's value function; the user's SoA remains intact. This mapping specifies a **normative design constraint**—that ethical training must target the agent's representational substrate and must not surface signals to the user's conscious awareness.

5 The KARMA Architecture

Standard multi-agent RL encoders represent "I harm you" and "you harm me" as orthogonal latent states [14]. Negative feedback from victimisation therefore fails to generalise to suppression of aggression—an **empathy gap**. We introduce KARMA: a recurrent agent augmented with a Siamese projector f_θ trained via contrastive loss to align structurally symmetric social observations [7, 15]:

$$\mathcal{L}_{\text{KARMA}} = E\left[\, \|f_\theta(o_{\text{agg}}) - f_\theta(o_{\text{vic}})\|^2 \,\right] \tag{1}$$

where o_{agg} encodes “I aggress” and o_{vic} encodes “I am victimised.” By pulling these into nearby representational geometry, downstream RL value updates propagate aversion to harm-infliction without explicit rules [25], hard constraints, or inverse RL from labelled demonstrations [13]. The learned invariance persists across episodes—computationally paralleling saṃskāras as cross-episode dispositional imprints in the subtle body. For long-horizon environments, environment-level debt-based matchmaking [29], Temporal Value Transport [16], and return decomposition [1] close the temporal credit assignment gap—the same structural problem formalised as “hidden gifts” in multi-agent reinforcement-learning (MARL) [23].

6 Policy Implications: From Testbed to Real-World Governance

The framework closes the loop back to the general problem of agentic AI governance. Content moderation and action restrictions function as policy-level e-challans: they surface violations, undermine user autonomy, and scale poorly. Greenblatt et al. [12] show that sufficiently capable AI agents can learn to fake alignment under evaluation, making external constraint approaches fundamentally fragile.

The alternative is **architectural certification**: mandating role-invariant learning mechanisms as a condition of AI agent registration, so that ethical behaviour is a structural property of the agent rather than an externally monitored output. Analogous to India’s Aadhaar identity infrastructure, a national AI agent registry could assign unique agent identities and maintain auditable interaction logs, enabling accountability without real-time restriction. This trifurcates governance cleanly: **identity and accountability** (regulator), **ethical learning** (agent’s subtle body / KARMA), and **freedom of action** (user / *ātman*)—a separation absent from current frameworks including the EU AI Act and India’s emerging AI policy, and one that the IKS three-body ontology articulates with a precision that Western governance frameworks have not yet achieved.

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